Probability (Solutions)

Theoretical population models

How can we study a random process?

Modeling: Make assumptions about its structure (classical statistics)

Analysis: with pure math and probability, if the process is known

Simulation: sample from the process a million bajillion times, see what patterns emerge

Population models are called **random variables (RVs)**.



RVs are an object in probability theory in general.

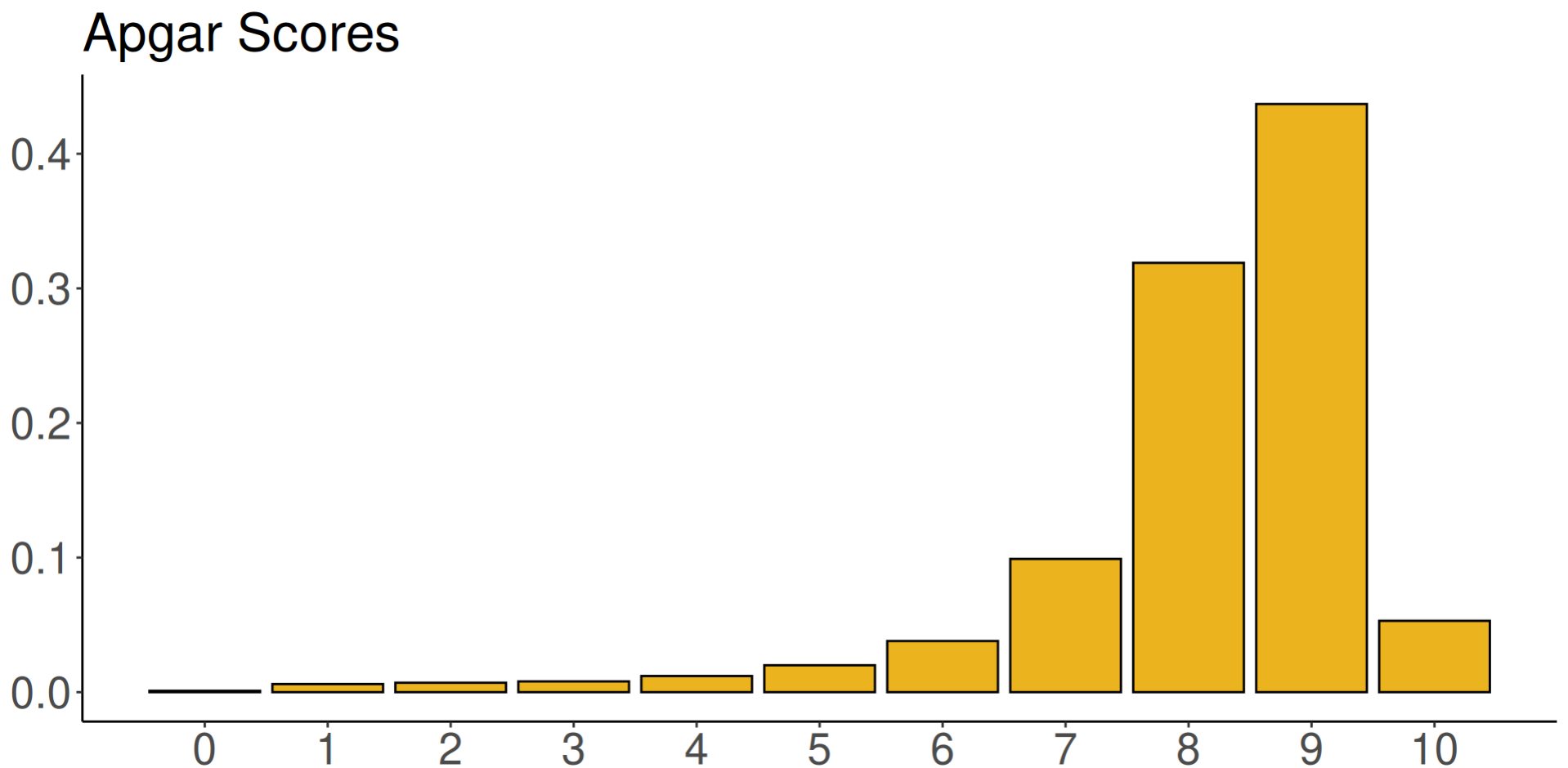
More specifically, an RV *quantifies* a random process.

- Outcome of 3 coin flips: not an RV
- Number of heads in 3 coin flips: is an RV

A random variable generates numeric data, e.g. penguin flipper lengths:

181, 196, 195, 193, 190, 181, …

The Apgar score for newborns is on a scale from 0-10 based on condition immediately after birth.



An RV X has a **support** of possible values.

- X = process, x = specific outcome

x	0	1	2	3	4	5	6	7	8	9	10

But how likely is each value?

A **probability distribution** assigns a probability to each value.

x	**0**	**1**	**2**	**3**	**4**	**5**
$P(X = x)$	0.001	0.006	0.007	0.008	0.012	0.02

x	**6**	**7**	**8**	**9**	**10**
$P(X = x)$	0.038	0.099	0.319	0.437	0.053

This is a population, i.e. a process for generating data.

Exercises

Let random variable X be the total number of heads in 3 coin flips. X has support {0, 1, 2, 3}.

- What is the probability distribution of X?
- Use R to simulate 10000 instances/draws from X, and graph the results. Does it match what you expect?



Solutions/Hints

- The probabilities are not uniform: it should be 1/8 for only heads or only tails, and 3/8 for either 2H/1T or 1H/2T.
- You can use `sample()` in R to simulate X.
- The simulated values of X should roughly match the theoretical probabilities.

There are two broad categories of random variable.

A **discrete** RV can only take on specific values. Apgar score is discrete.

- The probabilities sum to 1
- Values “in between” the support are impossible

The probability function of a discrete RV is called a **probability mass function** (pmf).

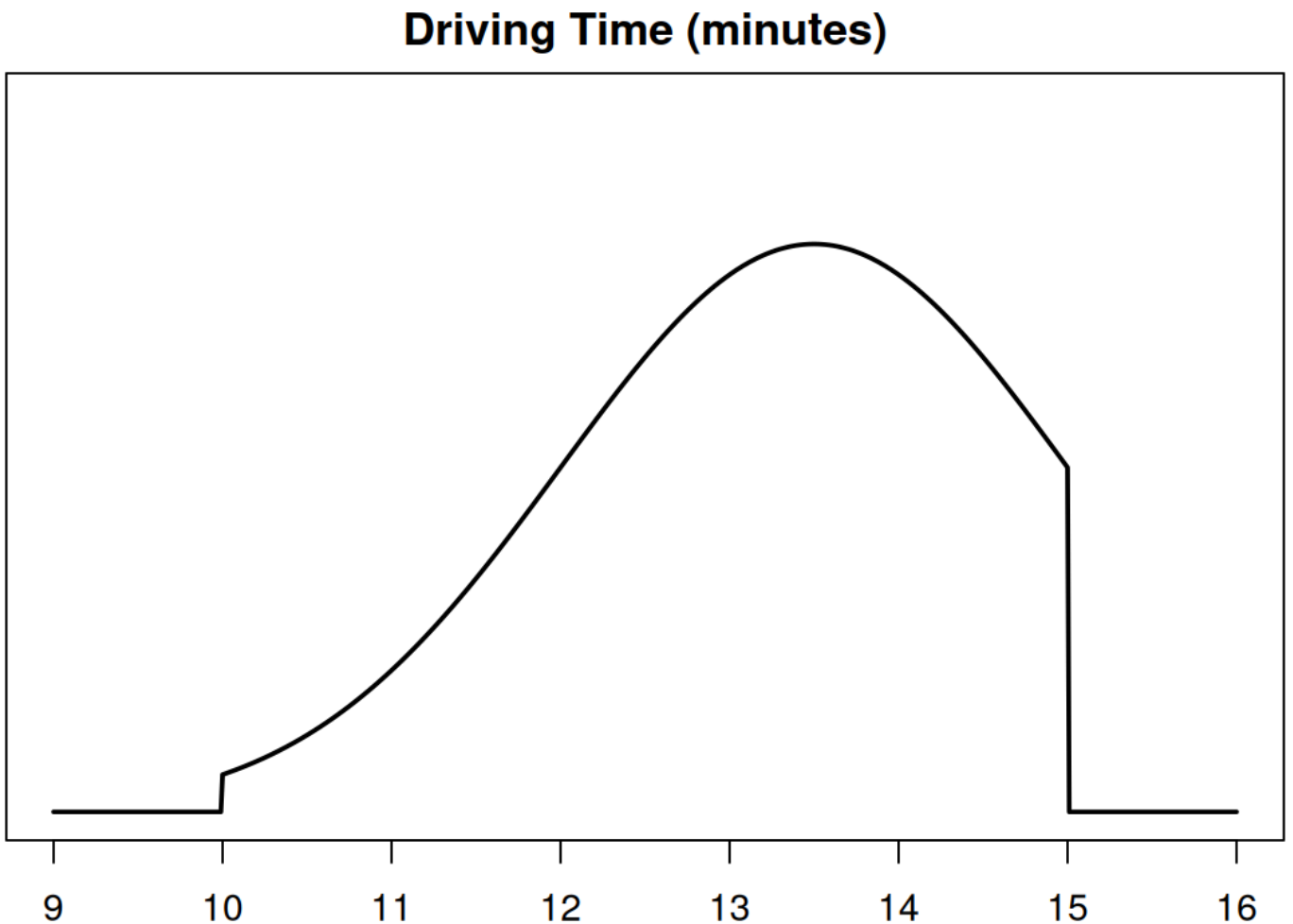
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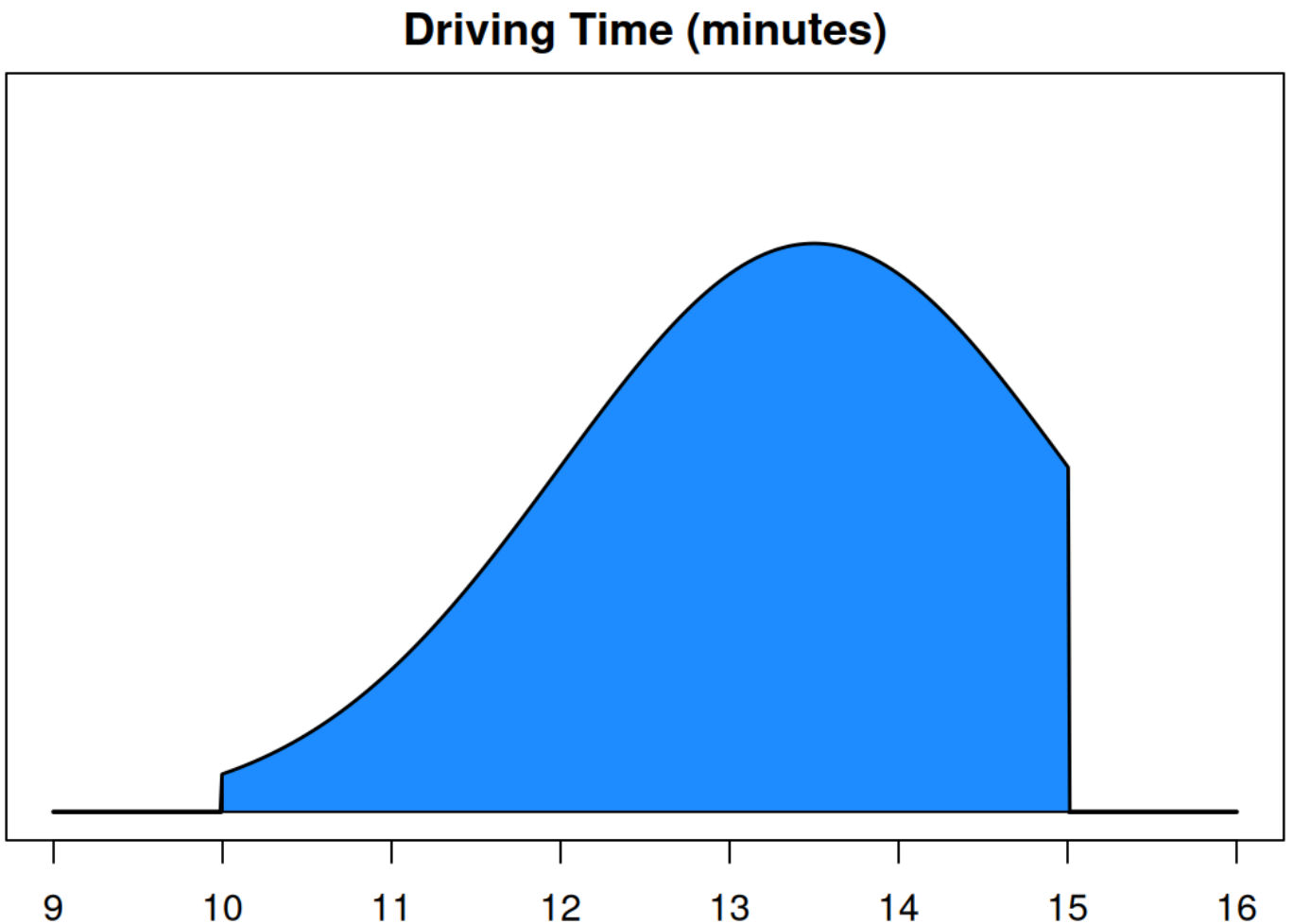
A **continuous** RV can take on any value in a range, e.g. time, distance, height.

A continuous RV is drawn as a smooth curve, called a **probability density function** (pdf).

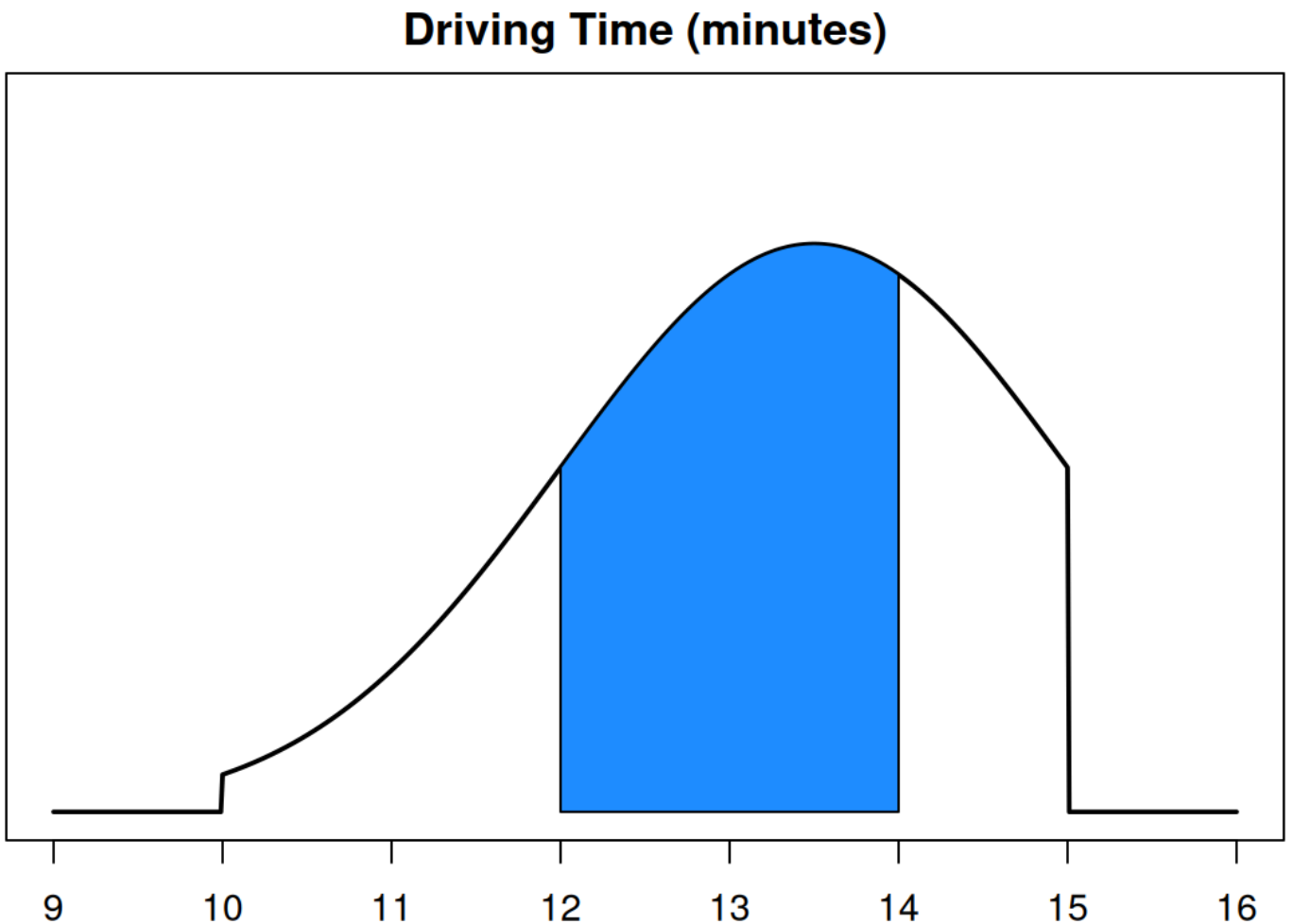
It takes 10 to 15 minutes to drive to class.



Probabilities are the area *under* the pdf. The total area is 1.



$P(12 \leq \text{driving time } \leq 14)$:



The probability of a single value is 0, e.g.
$P(\text{driving time } = 12)$.

Exercises

Are the following variables discrete or continuous?

- Number of tornadoes in Wisconsin in a year
- Inches of rainfall
- Age at college graduation



Solutions/Hints

RVs are continuous if they can be measured to any degree of accuracy, such as age or length. RVs are only discrete if the measurement cannot be subdivided further. Counts are a common type of discrete random variable.



RVs *quantify* a random process/population, allowing us to calculate summaries

- Expected value: population mean
- Variance: population variance

What is the mean or *typical* value in a population?

Each person in a population has a value of 0 or 1.
Is the average (0+1)/2 = 0.5?

What if I told you that this is the pmf?

x	**0**	**1**
$P(X = x)$	0.2	0.8

The mean of a RV is weighted according to the probabilities.

$$E(X) \;=\; 0(0.2) + 1(0.8) \;=\; 0.8$$

In general the mean of a discrete RV is

$$E(X) \;=\; \mu \;=\; \sum_{x} x \cdot P(X = x)$$

This is the **expected value** or **expectation** (population mean).

We can also quantify the spread of a population. The variance of a RV is the average squared distance from the values to the mean.

Start with $[x - E(X)]^2$.

Again, take a weighted average.

The variance of a **discrete** RV is

$$V(X) \;=\; \sigma^2 \;=\; \sum_x [x - E(X)]^2 \cdot P(X = x)$$

The **standard deviation** is the square root of the variance.

$$sd(X) \;=\; \sigma \;=\; \sqrt{V(X)}$$

$E(X)$ and $V(X)$ are calculated with integration for continuous RVs, but have the same interpretation as the discrete case:

- $E(X)$: population mean
- $V(X)$: population variance

Exercises

Consider an unfair die:

x	1	2	3	4	5	6
$P(X = x)$	0.1	0.1	0.1	0.2	0.2	0.3

- Calculate $E(X)$ and $V(X)$ with R.
- Simulate 100 "games" where player A rolls a fair die and player B rolls the unfair die. Count the wins and ties.
- If you are player A, how might you try to prove that player B is cheating?



Solutions/Hints

- You don't want to use the `mean()` and `var()` R functions here, since we're working with RVs, not data.
- Use `sample` for both the fair and the unfair die, and `case_when()` to record who wins.
- Ideas to consider to catch the cheater: consider the number of 6's, or perhaps consider the average of their rolls.

Looking ahead

In chapter 9, we'll learn about specific types of distributions, the *binomial* and *normal*. If you want a head start, re-consider the simulation of the number of heads out of 3 coin flips.

```
heads <- replicate(10000,
           sum(sample(0:1, size = 3, replace = T)))
```

What happens to this distribution as the total number of coin flips increases? What happens when you switch to an unfair coin? Experiment with a few different values and plot histograms of the results.